# Integration of Quantum Solvers with 3D & 4D Volumetric Characterization (ARPES, STEM, and More)

Integration of Quantum Solvers with 3D & 4D Volumetric Characterization by Onri Jay Benally

For high data volume approaches, focus on multi-slice simulation. Why? Because the data is holds considerable commercial, economic, and scientific value. Keyword: high-value asset

EXP = Experimental acquisition REC = Numerical reconstruction SIM = Forward numerical simulation HYB = Hybrid experimental-computational analysis

3D & 4D Volumetric Characterization & Reconstruction

- 1. Reciprocal-Space Spectroscopy, Scattering & Momentum Mapping
  - A. 3D ARPES Energy-Momentum Spectral Mapping [EXP + REC]
    - Fixed-Photon-Energy ARPES: I(kx, ky, E)
      - $ZrAs_2$ Poster Method (30 eV and 100 eV)
    - Tilt-Scan/ Deflector-Based ARPES
    - Time-of-Flight Momentum Microscopy (ToF-MM)
      - Parallel Momentum-Energy Acquisition
  - B. 4D ARPES Volumetric Band Mapping [EXP + REC]
    - Photon-Energy-Dependent ARPES
      - Measured Stack: I(kx, ky, hv, E)
    - Out-of-Plane Momentum (kz) Reconstruction
    - Reconstructed Volume: I(kx, ky, kz, E)
      - Bulk/Surface Electronic-State Analysis
  - C. 3D ARPES Fermi-Surface Tomography [EXP + REC]
    - Iso-Energy Mapping: I(kx, ky, kz) at E = EF
    - VUV/ Soft-X-Ray Photon-Energy Scanning
    - 3D Fermi-Surface Reconstruction
      - Electron and Hole Pockets
      - Nodal Lines and Fermi-Surface Topology
  - D. Photoemission Orbital Tomography (POT) [EXP + REC]
    - Molecular-Orbital Momentum Fingerprints
    - Fourier-Domain Phase Retrieval
    - Reciprocal-to-Real-Space Orbital Reconstruction
      - 3D Molecular-Orbital Visualization
  - E. 3D/4D Inelastic Neutron Scattering (INS) [EXP + REC]

- Single-Crystal Time-of-Flight Neutron Spectroscopy
  - Multi-Orientation Momentum-Energy Acquisition
    - 4D Dynamical Structure Factor S(Qx, Qy, Qz, ω)
  - 3D Momentum-Energy Spectral Mapping
    - S(Qx, Qy, ω) at Fixed Qz
  - 3D Constant-Energy Momentum Tomography
    - S(Qx, Qy, Qz) at Fixed Energy Transfer
  - Magnetic Excitation Spectroscopy
    - Magnons and Spin-Wave Dispersions
    - Spinon and Multiparticle Continua
    - Polarization-Resolved Spin Correlations
  - Lattice-Dynamical Spectroscopy
    - Phonon Dispersions and Polarizations
    - Anharmonic and Phonon-Coupling Effects
  - 4D Data Processing and Visualization
    - Momentum-Space Registration and Symmetrization
    - 3D Volumetric Intensity Visualization
    - 2D Momentum-Energy Dispersion Slices

- 2. High-Resolution 3D STEM & Atomic Reconstruction
  - A. 3D STEM Electron Tomography [EXP + REC]
    - Conventional Tilt-Series Electron Tomography
      - Bright-Field TEM (BF-TEM) Tomography
      - Annular Dark-Field STEM (ADF-STEM)
    - HAADF-STEM Tomography
      - 3D Morphology and Atomic Structure
    - Atomic Electron Tomography (AET)
      - 3D Atomic-Coordinate Reconstruction
      - Atomic Defect and Vacancy Mapping
      - 3D Lattice and Structural Distortions
  - B. 3D Diffraction-Resolved STEM Tomography [EXP + REC]
    - 4D-STEM Diffraction Acquisition
      - I(x, y, qx, qy) Across Specimen Tilts
    - Ptychographic Atomic Electron Tomography (pAET)
      - 3D Atomic-Structure Reconstruction
      - 3D Phase/ Electrostatic-Potential Contrast
    - 3D Differential Phase Contrast STEM (DPC-STEM)
      - 3D Electric-Field Reconstruction
      - Electrostatic-Potential Reconstruction
    - Scanning Precession Electron Diffraction (SPED)
      - 3D Crystallographic Orientation Mapping
      - Grain and Phase-Structure Reconstruction
  - C. 3D Depth-Resolved STEM Ptychography [EXP + REC]
    - Multislice Electron Ptychography (MEP)
      - Depth-Resolved Atomic-Potential Slices

- Multislice Electron-Wave Reconstruction
- 3D Atomic-Column and Interface Analysis
- Tilt-Coupled Multislice Ptychography (TC-MEP)
  - Multi-Tilt 4D-STEM Acquisition
  - Improved Axial/ Depth Resolution
  - 3D Atomic-Feature Reconstruction
  - Individual Dopant and Defect Localization

- 3. Computational Electronic Structure, Wave Simulation & 3D Reconstruction
  - A. DFT-Based Electronic-Structure Prediction [SIM]
    - DFT/ DFT+SOC Band Structures
      - Compare with 1A: 3D ARPES Spectral Mapping
    - 3D Brillouin-Zone Band-Energy Volumes
      - Compare with 1B: 4D ARPES Volumetric Mapping
    - Wannier-Interpolated 3D Fermi Surfaces
      - Compare with 1B/1C: 4D ARPES/ 3D FS Tomography
    - DFT-Relaxed 3D Atomic & Defect Structures
      - Compare with 2A: 3D Atomic Tomography
    - 3D Charge-Density and Electrostatic Potentials
      - Compare with 2B/2C: DPC-STEM/ Ptychography
    - DFPT Phonon Dispersion and Vibrational Modes
      - Compare with 1E: Lattice-Dynamical INS Spectroscopy
  - B. Beyond-DFT Correlated Spectral Predictions [SIM]
    - GW Quasiparticle Corrections
    - DFT+DMFT and Quantum Impurity Solvers
    - Many-Body Green's Functions
    - Correlated Spectral Functions A(k, ω)
      - Compare with 1A/1B/1C: Experimental ARPES
    - Dynamical Spin Correlations and Many-Body Spectra
      - Compare with 1E: Magnetic Excitation INS Spectroscopy
    - QPU-Assisted Many-Body Hamiltonian Simulation
  - C. Synthetic 3D/4D ARPES Forward Modeling [SIM]
    - Synthetic Spectral Volumes: I(kx, ky, E)
      - Compare with 1A: Experimental 3D ARPES
    - Synthetic 4D Volumes: I(kx, ky, kz, E)
      - Compare with 1B: Experimental 4D ARPES
    - 3D Fermi-Surface Intensity Reconstruction
      - Compare with 1C: 3D FS Tomography
    - Photoemission Matrix-Element Calculations
    - Molecular-Orbital Photoemission Modeling
      - Compare with 1D: Photoemission Orbital Tomography
  - D. Atomistic Structure & Electrostatic Modeling [SIM]
    - DFT-Relaxed 3D Atomic Coordinates
      - Compare with 2A: AET/ HAADF-STEM

- 3D Atomic Strain and Defect Distributions
  - Compare with 2A: Atomic Tomography
- 3D Electrostatic Potential Volumes V(x, y, z)
  - Compare with 2B: 3D DPC/ pAET
- Charge Redistribution and Bonding Potentials
  - Compare with 2B/2C: STEM Ptychography

- E. Computational 3D Atomic Tomography [SIM + REC]
  - Fourier-Based Iterative Reconstruction
    - GENFIRE/ Fourier-Slice Algorithms
  - Real-Space Iterative Reconstruction
    - RESIRE/ Radon-Operator Inversion
  - Algebraic Tomographic Reconstruction
    - ART/ SIRT/ SART
  - 3D Atomic-Coordinate and Defect Extraction
    - Compare with 2A: Experimental AET/ HAADF-STEM

- F. Multislice Electron-Wave & STEM Forward Modeling [SIM]
  - Paraxial Schrödinger-Wave Propagation
  - Split-Operator Multislice Propagation
  - PRISM Scattering-Matrix Interpolation
  - Synthetic HAADF-STEM Tilt Series
    - Compare with 2A: STEM Tomography
  - Synthetic 4D-STEM Diffraction Patterns
    - Compare with 2B: Diffraction Tomography
  - Synthetic DPC-STEM/ Ptychographic Phase
    - Compare with 2B: DPC/ pAET
  - Depth-Resolved Electron-Wave & Potential Volumes
    - Compare with 2C: MEP/ TC-MEP

- G. Inverse Multislice & 3D Ptychography [REC]
  - Iterative Multislice Phase Retrieval
  - Differentiable Ptychographic Reconstruction
  - Ptychographic Atomic Electron Tomography
    - Compare with 2B: Experimental pAET
  - Tilt-Coupled Multislice Reconstruction
    - Compare with 2C: Experimental TC-MEP
  - Electrostatic-Potential Volume Estimation
    - Compare with 2B/2C: DPC/ MEP
  - Atomic Dopant/ Defect Localization
    - Compare with 2A/2C: AET/ TC-MEP

- H. Quantum-Compatible Wave & PDE Solvers [SIM]
  - Quantum Hamiltonian Simulation
    - Electron-Wave Propagation Operators
    - Interacting Electronic Hamiltonians
    - Dynamical Spin-Correlation Functions
  - Quantum FDTD/ Maxwell Hamiltonian Solvers

- Electromagnetic Field Evolution
- Schrödingerisation
  - Linear Nonunitary PDE Embedding
- Quantum Linear-System/ QSVT Methods
  - Encoded Linear Inverse Problems
- Wave Matrix Lindbladization
  - Open-System Quantum Dynamics
- Ehrenfestization
  - Reduced Quantum Observable Dynamics

- I. Theory-to-Experiment Registration & Benchmarking [HYB]
  - 3D/4D ARPES Spectral-Volume Registration
    - Compare with 1A/1B/1C: Experimental ARPES Volumes
  - Electronic Band and Self-Energy Fitting
  - 3D Atomic-Coordinate and Defect Comparison
    - Compare with 2A/2C: AET/ TC-MEP Coordinates
  - Electrostatic-Potential Volume Registration
    - Compare with 2B/2C: DPC/ pAET/ MEP Potentials
  - Synthetic/ Experimental Diffraction Matching
    - Compare with 2B: 4D-STEM Diffraction Stacks
  - Synthetic/ Experimental INS Volume Registration
    - Compare with 1E: 3D/4D Neutron Spectra
  - Convergence, Uncertainty, and Reconstruction Fidelity
  - CPU/ GPU/ QPU Resource Benchmarking
  - Reproducible 3D/4D Scientific Visualization

- J. Synthetic Inelastic Neutron Scattering [SIM]
  - 3D/4D Dynamical Structure Factors
    - S(Qx, Qy, Qz, ω)
  - First-Principles Phonon Scattering
    - DFT/ Density-Functional Perturbation Theory
    - Phonon Eigenmodes and Polarizations
    - Synthetic Vibrational INS Intensities
  - Magnetic Neutron-Scattering Simulation
    - Linear Spin-Wave Theory (LSWT)
    - SpinW and Magnetic Scattering Cross Sections
    - Exact Diagonalization/ Tensor Networks
    - Quantum Spin-Spin Correlation Functions
  - QPU-Native Dynamical Structure Factors
    - Spin Hamiltonian State Preparation
    - Quantum Real-Time Evolution
    - Dynamical Correlation Measurements
    - Fourier Reconstruction of S(Q, ω)
  - Experimental Neutron-Spectrum Validation
    - Phonon and Magnon Dispersion Comparison
    - Spectral Intensity and Linewidth Fitting
    - Compare with 1E: Experimental 3D/4D INS